

SCIENCE OF MUSCLE HYPERTROPHY

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Ch 1: The Three Mechanisms

Current evidence identifies three primary stimuli driving muscle protein synthesis and hypertrophy: mechanical tension, metabolic stress, and muscle damage. Understanding each allows programme design that maximises all three.

Mechanism	Description	How to Maximise	Optimal Rep Range
Mechanical Tension	Force applied to sarcomeres during contraction and stretch	Heavy compound lifts with full ROM; slow controlled eccentrics	3–8 reps (85–90% 1RM)
Metabolic Stress	Accumulation of lactate, inorganic phosphate, and H ⁺ ions	Higher reps with short rest; occlusion/BFR training	12–20 reps (60–75% 1RM)
Muscle Damage	Micro-tears from eccentric and novel loading patterns	Novel exercises; lengthened partials; 3–5 sec eccentric tempo	8–15 reps with slow eccentric

Ch 2: Evidence-Based Training Variables

Variable	Hypertrophy Optimum	Key Research Finding
Sets per muscle per week	10–20 sets (MEV to MRV)	Start at 10; add 1 set/muscle/week until plateau; deload then re-establish MV
Reps per set	6–20 (peak stimulus: 8–12)	Multiple rep ranges within 6–20 produce fuller hypertrophy stimulus
Effort (RIR)	0–2 reps in reserve per set	Proximity to failure is the strongest predictor of hypertrophy stimulus
Rest between sets	60–180 seconds	Compound lifts: 2–3 min; isolation: 60–90 sec for metabolic effect
Training frequency	2–3x/week per muscle group	More MPS spikes per week with higher frequency; same total volume
Eccentric tempo	2–4 seconds	Slow eccentrics enhance mechanical tension mechanism
Range of motion	Full ROM; emphasise stretched position	Lengthened partials show surprisingly high EMG and hypertrophy

Ch 3: Progressive Overload Methods

Method	Application	Best Phase	Example
Load progression	Add 2.5–5 kg when rep target achieved	Strength base phase	Bench: 80 kg x 4 → 82.5 kg x 4 next session
Volume progression	Add 1 set per exercise per week	Hypertrophy phase	3x10 → 4x10 → 5x10 over 3 weeks
Density progression	Same total work in less time	Metabolic/conditioning phase	Same sets/ reps; rest drops 90→75→60 sec over weeks
Frequency progression	Add training days per muscle group	Advanced phase	3x/week → 4x/week upper body as volume tolerance increases
ROM progression	Increase range of motion over time	Skill/technique phase	Squat depth improvements; fuller pull-up ROM
Technique progression	Reduce compensations; purer muscle activation	Foundation phase	Eliminate bar drift in deadlift; lat engagement in rows

Ch 4: Muscle Protein Synthesis (MPS)

MPS is the molecular process by which the body repairs and builds new muscle proteins. A single training session elevates MPS for 24–48 hours. For hypertrophy, MPS must consistently exceed muscle protein breakdown (MPB). Nutrition is the primary lever.

MPS Driver	Requirement	Practical Application
Leucine threshold	~2.5–3g leucine per meal to trigger MPS	40g whey = 3.5g leucine; 200g chicken = 3g leucine
Protein per meal	0.4 g/kg body weight per meal	80 kg person = 32g protein every 4–5 hours
Daily protein total	1.6–2.2 g/kg BW/day	80 kg person = 128–176g total protein daily
Protein distribution	Even spread across 4–5 meals	Not front- or back-loaded into 1–2 large meals
Pre-sleep protein	40g casein/cottage cheese before bed	Maintains MPS elevation during overnight fast

Ch 5: Periodisation & Sample Programme

Phase	Duration	Rep Range	Intensity	Primary Goal
Anatomical adaptation	2–4 weeks	15–20	60–65%	Tendon/ligament preparation; form mastery
Hypertrophy (accumulation)	6–8 weeks	8–12	70–80%	Maximum muscle growth via volume accumulation
Strength (intensification)	4–6 weeks	3–6	82–92%	Neural efficiency; strength base for future hypertrophy
Power (optional)	2–4 weeks	3–5 explosive	70–80%	Rate of force development; athletic performance
Deload	1 week	Reduce 50% volume	Reduce 30% load	Recovery; supercompensation; avoid overreaching

4-Day Upper/Lower Hypertrophy Template

Day	Session	Priority Exercises	Sets x Reps
Monday	Upper (strength focus)	Bench press 4x5, Barbell row 4x5, OHP 3x6, Weighted pull-up 3x5	Heavy; 2–3 min rest
Tuesday	Lower (strength focus)	Squat 4x5, Romanian DL 3x6, Leg press 3x8, Leg curl 3x8	Heavy; 3 min rest
Thursday	Upper (hypertrophy focus)	Incline DB press 4x10, Cable row 3x12, DB shoulder press 3x12, Curl 3x12	Moderate; 75 sec rest
Friday	Lower (hypertrophy focus)	Leg press 4x12, Bulgarian split squat 3x10, Hack squat 3x12, Glute bridge 3x15	Moderate; 90 sec rest